

Dissertation Preparation

Literature Review by Alexandre Valle and Daniel Mendes
v20260928

Dissertation Contracts

Instructions will follow in the next weeks.

Os modelos de contrato e de NDA/acordo de confidencialidade estão disponíveis no site de Dissertações do M.EIC apenas para consulta e referência. O pedido de emissão de contrato e de NDA deve ser feito através do preenchimento do formulário na plataforma M.EIC Thesis, conforme indicado no e-mail que deve ter recebido dos regentes de Dissertação. De notar que não são aceites contatos ou acordos que não tenham sido emitidos pelo secretariado e previamente assinados pelo Diretor do M.EIC.

Previously, on 2nd class...

Table 1 Main review types characterized by methods used

Label	Description	Methods used (SALSA)			
		Search	Appraisal	Synthesis	Analysis
Critical review	Aims to demonstrate writer has extensively researched literature and critically evaluated its quality. Goes beyond mere description to include degree of analysis and conceptual innovation. Typically results in hypothesis or model	Seeks to identify most significant items in the field	No formal quality assessment. Attempts to evaluate according to contribution	Typically narrative, perhaps conceptual or chronological	Significant component: seeks to identify conceptual contribution to embody existing or derive new theory
Literature review	Generic term: published materials that provide examination of recent or current literature. Can cover wide range of subjects at various levels of completeness and comprehensiveness. May include research findings	May or may not include comprehensive searching	May or may not include quality assessment	Typically narrative	Analysis may be chronological, conceptual, thematic, etc.
Mapping review/ systematic map	Map out and categorize existing literature from which to commission further reviews and/or primary research by identifying gaps in research literature	Completeness of searching determined by time/scope constraints	No formal quality assessment	May be graphical and tabular	Characterizes quantity and quality of literature, perhaps by study design and other key features. May identify need for primary or secondary research
Meta-analysis	Technique that statistically combines the results of quantitative studies to provide a more precise effect of the results	Aims for exhaustive, comprehensive searching. May use funnel plot to assess completeness	Quality assessment may determine inclusion/exclusion and/or sensitivity analyses	Graphical and tabular with narrative commentary	Numerical analysis of measures of effect assuming absence of heterogeneity
Mixed studies review/mixed methods review	Refers to any combination of methods where one significant component is a literature review (usually systematic). Within a review context it refers to a combination of review approaches for example combining quantitative with qualitative research or outcome with process studies	Requires either very sensitive search to retrieve all studies or separately conceived quantitative and qualitative strategies	Requires either a generic appraisal instrument or separate appraisal processes with corresponding checklists	Typically both components will be presented as narrative and in tables. May also employ graphical means of integrating quantitative and qualitative studies	Analysis may characterise both literatures and look for correlations between characteristics or use gap analysis to identify aspects absent in one literature but missing in the other
Overview	Generic term: summary of the [medical] literature that attempts to survey the literature and describe its characteristics	May or may not include comprehensive searching (depends whether systematic overview or not)	May or may not include quality assessment (depends whether systematic overview or not)	Synthesis depends on whether systematic or not. Typically narrative but may include tabular features	Analysis may be chronological, conceptual, thematic, etc.
Qualitative systematic review/qualitative evidence synthesis	Method for integrating or comparing the findings from qualitative studies. It looks for 'themes' or 'constructs' that lie in or across individual qualitative studies	May employ selective or purposive sampling	Quality assessment typically used to mediate messages not for inclusion/exclusion	Qualitative, narrative synthesis	Thematic analysis, may include conceptual models

A typology of reviews: an analysis of 14 review types and associated methodologies, Maria J. Grant* & Andrew Booth†, *Salford Centre for Nursing, Midwifery and Collaborative Research (SCNMCR), University of Salford, Salford, UK, †School of Health and Related Research (ScHARR), University of Sheffield, Sheffield, UK

[DOI: 10.1111/j.1471-1842.2009.00848.x](https://doi.org/10.1111/j.1471-1842.2009.00848.x)

Table 1 *Continued*

Label	Description	Methods used (SALSA)			
		Search	Appraisal	Synthesis	Analysis
Rapid review	Assessment of what is already known about a policy or practice issue, by using systematic review methods to search and critically appraise existing research	Completeness of searching determined by time constraints	Time-limited formal quality assessment	Typically narrative and tabular	Quantities of literature and overall quality/direction of effect of literature
Scoping review	Preliminary assessment of potential size and scope of available research literature. Aims to identify nature and extent of research evidence (usually including ongoing research)	Completeness of searching determined by time/scope constraints. May include research in progress	No formal quality assessment	Typically tabular with some narrative commentary	Characterizes quantity and quality of literature, perhaps by study design and other key features. Attempts to specify a viable review
State-of-the-art review	Tend to address more current matters in contrast to other combined retrospective and current approaches. May offer new perspectives on issue or point out area for further research	Aims for comprehensive searching of current literature	No formal quality assessment	Typically narrative, may have tabular accompaniment	Current state of knowledge and priorities for future investigation and research
Systematic review	Seeks to systematically search for, appraise and synthesis research evidence, often adhering to guidelines on the conduct of a review	Aims for exhaustive, comprehensive searching	Quality assessment may determine inclusion/exclusion	Typically narrative with tabular accompaniment	What is known; recommendations for practice. What remains unknown; uncertainty around findings, recommendations for future research
Systematic search and review	Combines strengths of critical review with a comprehensive search process. Typically addresses broad questions to produce 'best evidence synthesis'	Aims for exhaustive, comprehensive searching	May or may not include quality assessment	Minimal narrative, tabular summary of studies	What is known; recommendations for practice. Limitations
Systematized review	Attempt to include elements of systematic review process while stopping short of systematic review. Typically conducted as postgraduate student assignment	May or may not include comprehensive searching	May or may not include quality assessment	Typically narrative with tabular accompaniment	What is known; uncertainty around findings; limitations of methodology
Umbrella review	Specifically refers to review compiling evidence from multiple reviews into one accessible and usable document. Focuses on broad condition or problem for which there are competing interventions and highlights reviews that address these interventions and their results	Identification of component reviews, but no search for primary studies	Quality assessment of studies within component reviews and/or of reviews themselves	Graphical and tabular with narrative commentary	What is known; recommendations for practice. What remains unknown; recommendations for future research

Literature Review

“comprehensive accumulation, transparent analysis and reflective interpretation of all empirical studies pertinent to a specific question”

Denise Rousseau, Joshua Manning, David Denyer, *Evidence in Management and Organisational Science: Assembling the Field's Full Weight of Scientific Knowledge through Syntheses* (August 1, 2008). Advanced Institute of Management Research Paper No. 067, <http://dx.doi.org/10.2139/ssrn.1309606>.

The trigger...

Domain and Context

Problem

Research question(s)

● Systematic? Not systematic ? Other?

Read: Sutton, A., Clowes, M., Preston, L. and Booth, A. (2019), *Meeting the review family: exploring review types and associated information retrieval requirements*.

Health Info Libr J, 36: 202-222. <https://doi.org/10.1111/hir.12276>

Relevant factors for strategy selection are:

- speed
- replicability
- comprehensibility
- bias

Narrative/exploratory review or Systematic review (and meta-analysis)? Or in-between? Discuss strategy with your supervisor!

- PRISMA method

PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) methodology.

<https://www.prisma-statement.org/>

<https://www.prisma-statement.org/prisma-2020-checklist>

<https://www.prisma-statement.org/prisma-2020-flow-diagram>



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	
Study characteristics	17	Cite each included study and present its characteristics.	
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	
	23b	Discuss any limitations of the evidence included in the review.	
	23c	Discuss any limitations of the review processes used.	
	23d	Discuss implications of the results for practice, policy, and future research.	
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	
Competing interests	26	Declare any competing interests of review authors.	
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71. doi: 10.1136/bmj.n71

<https://www.prisma-statement.org/prisma-2020-checklist>

Identification, screening, eligibility, inclusion

● Data sources and Search strings

Data sources: IEEE Xplore, SCOPUS, ScienceDirect, Clarivate Web of Science, ACM Digital Library, Google Scholar (this data source may be excluded because it is too broad), etc.

Search strings formulation and refinement:

- Start with one database
- Study search terms and search expression(s)
- run and inspect the search results
- understand if terms are too broad or are missing topics
- iterate!
- test with other databases!

● From Research Questions to Search terms

Before searching, decompose each review research question into structured concepts. This is what turns a question into a reproducible search string.

PICO originates in evidence-based medicine; **PICOC** adds **Context**, which is well-suited to software engineering (Petersen et al., 2015; Kitchenham & Charters, 2007).

P — Population: the group, technology, or artefact under study (e.g. developers, a language, a class of systems).

I — Intervention: the technique, tool, or method being examined.

C — Comparison: the alternative it is compared against (may be none for exploratory reviews).

O — Outcome: the effect or property measured (performance, usability, correctness...).

C — Context: the setting (academia vs industry, domain, scale).

Note: not every facet applies to every review

● PICOC worked example (illustration from [2])

Mapping the graph-visualisation review question onto PICOC:

Facet	Value
Population	Highly interconnected, time-evolving, multidimensional/multivariate data
Intervention	Graph-/network-based visualization techniques
Comparison	Other information-visualization approaches (no strict comparator)
Outcome	Ability to convey relationships while preserving the "story" over time
Context	Information visualization; excluding VR/AR/MR

Each facet feeds the search string. For example, *Population* → ("highly connected" OR interconnected) AND (data OR information) and *Intervention* → (graph OR network) ... visuali?ation, as in QS1 and QS4 (slide 14, class 2).

● From PICOC to search strings

1. List synonyms and spelling variants for each facet (e.g. *visualization* / *visualisation* → visuali?ation).
2. Join synonyms **within** a facet with **OR**.
3. Join **across** facets with **AND**.
4. Test on one database first, inspect results, then iterate (slide 13).
5. Adapt syntax per database — wildcards and proximity operators differ (WoS NEAR/n, Scopus W/n, IEEE rules).

(P-terms) AND (I-terms) AND (O-terms)

Keep the PICOC table in your protocol. If a reviewer asks, “*Why these terms?*”, the table is your

● Search strings (illustration from [2])

Last stage of search expressions:

QS1. (“highly connected” OR interconnected) AND (data OR information) AND visuali?ation

QS2. (time-evolving OR time-varying) AND (data OR information) AND visuali?ation

QS3. (multidimensional AND multivariate) AND (data OR information) AND visuali?ation

QS4. (graph OR network) AND ((time NEAR/2 evolving NEAR/2 data) OR (temporal NEAR/2 data)) AND visuali?ation

QS5. dynamic NEAR/1 graph NEAR/1 visuali?ation

Note: the NEAR/n and ? syntax is specific to Web of Science. Scopus uses W/n and PRE/n, and IEEE uses NEAR/n with different wildcard rules.

● Search strings justification (illustration from [2])

“The search strings consider the different spellings of the word “visualization” in British and American versions of English by using the wildcard operator.

The terms “data” or “information” and “visualization” by themselves appear in various contexts and domains.

The motivation of the first, second, and third search strings is to narrow down the search results by individually specifying the data characteristics inherent to the review (not excluding the possibility of obtaining articles that simultaneously present data with both traits).

The motivation of the fourth search string is to direct the search to results more related to the specific approach being explored.”

● Considerations

- **Carefully selecting and refining the search strings** is essential for finding relevant results, **saving time in the long run**.
- **Consider additional angles within the PRISMA framework** for extra references to address potential gaps in the search results.

This may require including renowned publications in the area, as referenced in other contexts, or exploring citations within the selected records.

Some publications that you are aware of, that are critical, and that were not identified in the sources

● Inclusion criteria (illustration from [2])

Adapted according to the capabilities of each data source:

- Journal articles, conference papers, and books.
- Recent publications (2018+).
- In the topic of “data visualization” or “graph theory” and area of “Computer Science” or “Engineering”.

	QS1	QS2	QS3	QS4	QS5	Total
IEEE	49	96	7	63	8	223
Scopus	164	232	39	133	26	602
ScienceDirect	1	7	0	2	1	11
WoS	121	184	42	70	18	435
ACM	18	10	2	-----		30

You will know the total number of papers that
were retrieved.

Inclusion and exclusion criteria (illustration from [2])

Phase s	Inclusion/Exclusion criteria
1	Search engine results after executing a search query
2	Date of the publication since 2018 (inclusive)
3	Full-text access
4	Published in the English language
5	Published in the area of “Information Visualization” or “Graph-based Visualization”
6	Not in the field of Virtual Reality (VR), Augmented Reality (AR), or Mixed Reality
7	Visualization of interconnected, temporal dependent, multi-dimensional, and/or multivariate information (share similar characteristics with the problem: time-evolving relationships)
8	Not duplicated
9	Not retracted publication
10	Not work-in-progress publication
11	Article, Conference Paper or Book
12	Not summaries of workshops, seminars, lectures, or tutorials

Explain and justify the inclusion/exclusion criteria

Screening process & results (illustration from [2])

Example: Starting from the **1301 records identified** in the previous stage.

Remove Duplicates:

- Performed using the EndNote capabilities in 3 phases: match of authors, year, title, and reference type; exact match of DOI; partial match of the title.
- Results in a dataset of **783 unique papers** and **518 duplicates**.

For screened records:

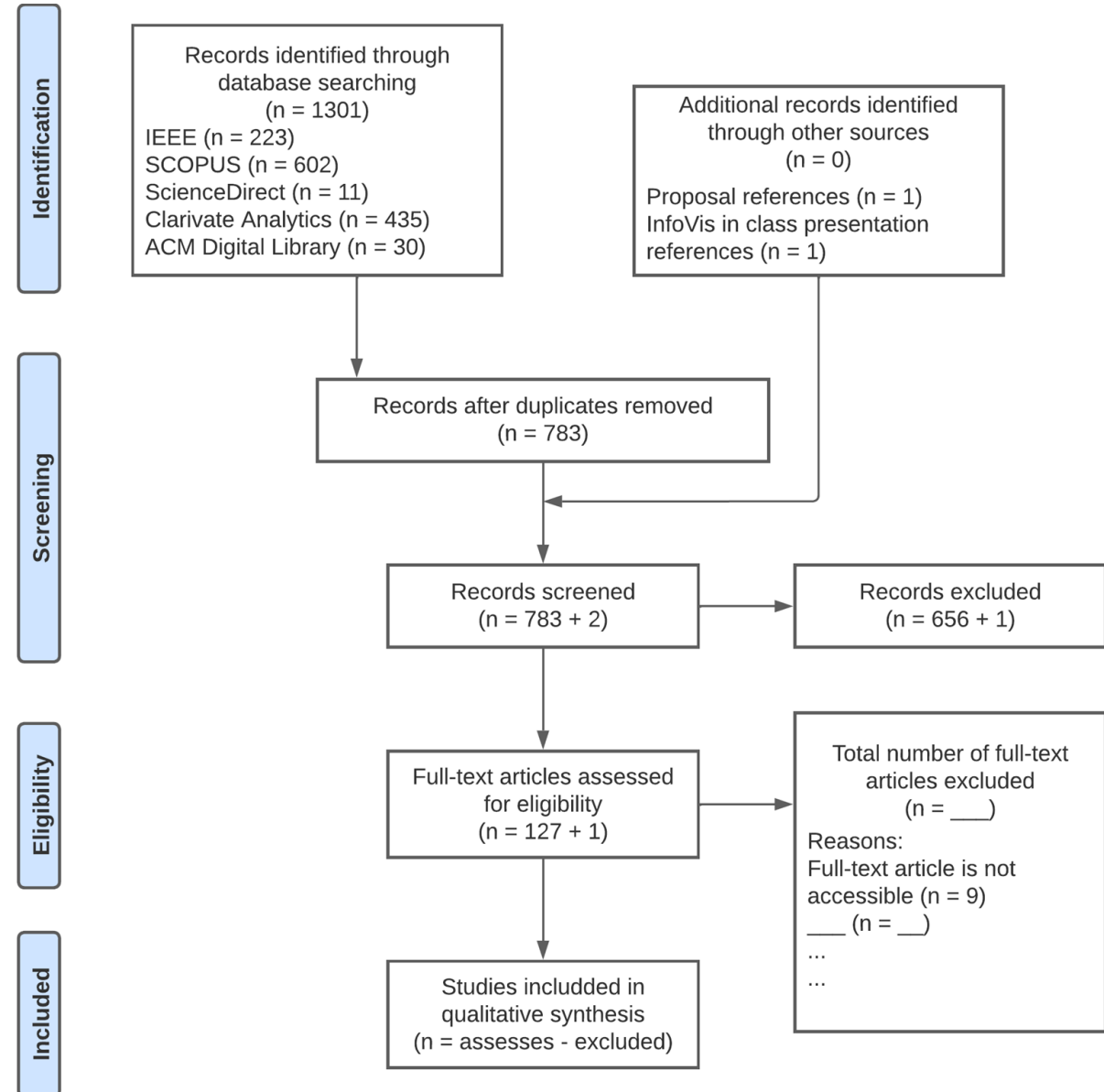
- Inspected the title and abstract of the 783 unique records to exclude the ones that do not match some criteria (reasoning annotated in the EndNote).
- Results in the **exclusion of 656** records, remaining **127 papers**.

▼	PRISMA	
👤	_ALL	783
📁	ACM	30
📁	Clarivate WoS	406
📁	IEEE	71
📁	ScienceDirect	8
📁	Scopus	268
▼	MY TAGS	+
📁	Excluded	656
📁	Maybe	
📁	Not Legible	9
📁	Promising	10
📁	Screened	127

Eligibility (illustration from [2])

Full-text eligibility:

- Used EndNote “full-text search” (located 26 PDFs, 70 URLs, 31 not found).
- Manually located the remaining PDFs.
- Identified **9 records** without full-text access.
- Verified if the year or paper type was mislabeled.
- Analysis of the full-text content:
 - Summarize research purpose and approach/domain.
 - Evaluate the relevance and quality contribution.
 - Save segments to potential cite in SOTA sections.
 - Reason about the contributions for the solution.



Inclusion...

1. **Read title**
2. **Read abstract**
3. **Read introduction**
4. **Read conclusions**
5. **Full-text read. It may be inevitable if the article is REALLY interesting.**

For each step (1..4) if you determine the paper is not interesting, exclude it and annotate: a) the process stage and b) the exclusion reason.

While you are full-text reading... (continue on the next slides please)

Content assessment and analysis

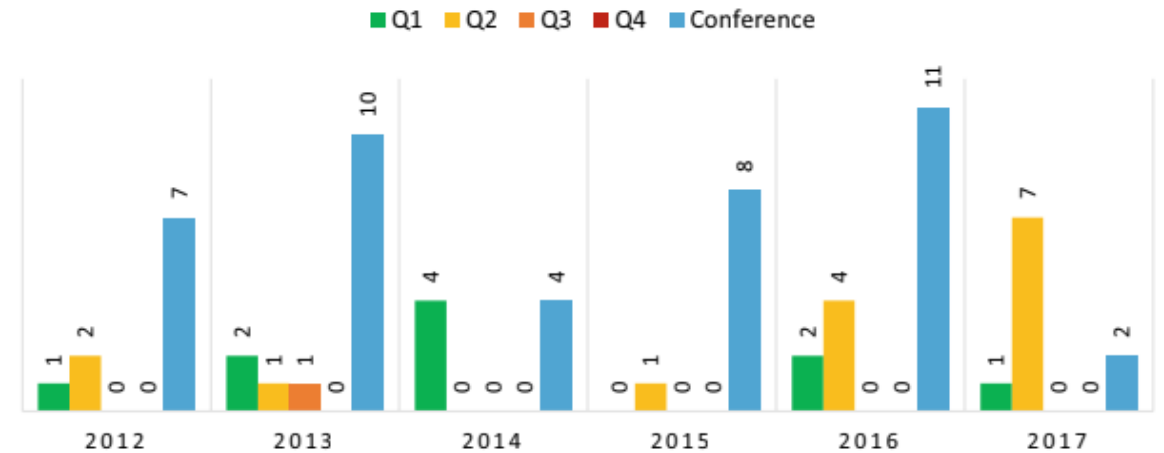
Content assessment and analysis

For each article read, annotate:

- Year
- Paper quartile (Q1, Q2, Q3, Q4, conference)

This will allow the creation of a chart and a discussion of the chart.

For instance, the subject's temporal development.



Example from [1]

Content assessment and analysis

For each article read, annotate:

- Main application domain/area
- Publication journal/conference

This will allow the creation of a chart and a discussion of the chart. For instance, the main application areas of the work read....

Publication journals/conferences allow identifying the main publication places

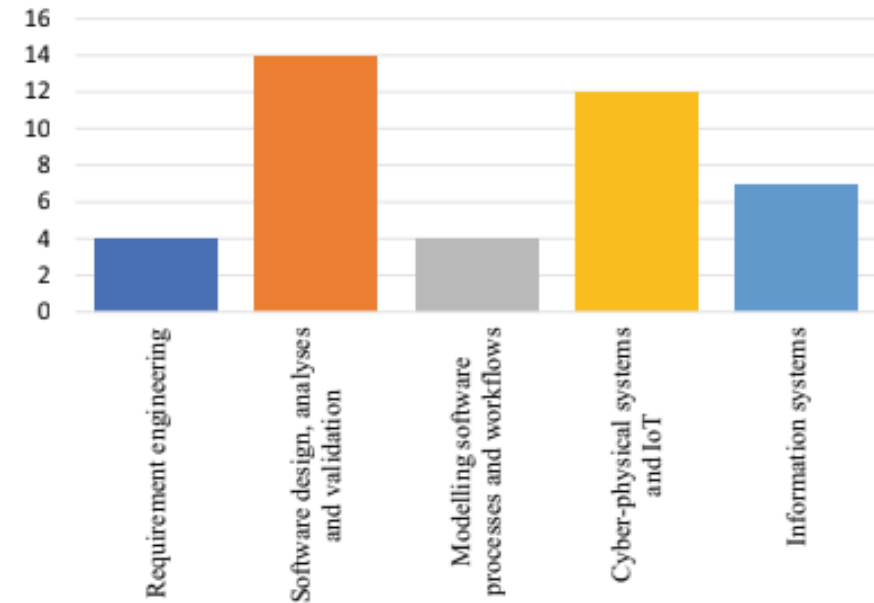


Fig. 3 The main application domains of metamodelling

Example from [1]

Content assessment and analysis

For each article read, annotate:

- Why was the work done (**motivation/problem**)?
- What was done (solution)?
- How was it validated (experiment)?
- What was concluded (results/conclusions)?
- Limitations?
- Trends for future research?

This will allow the creation of a table and discussion of it. The illustration on the right shows only 2 columns (but 6 columns are suggested)

Table 5 Research purpose description for the studies published since 2017

References	Research purpose	Approach/domain
[15]	To develop a framework that supports the behavior and structural complexity of combat system simulation	Simulation and software validation
[16]	To enhance the semantics of UML templates in OCL through the development of an aspectual template	Formalization, constraints, and software development
[17]	To simplify the development of domain-specific modeling tools	Code generation and software development
[18]	To develop the UML profile for modeling cognitive behavior	Development, cognitive behavior, and UML
[19]	To develop a flexible tool-supported modeling approach that augments a sketching environment with lightweight metamodeling capabilities	Development and sketching environment

Example from [1]

Content assessment and analysis

For each article, annotate:

- Which tools used?
- Which datasets used?

As said, each of these assessment items (and from the previous slide) allows the creation of visual charts/tables and a corresponding discussion.

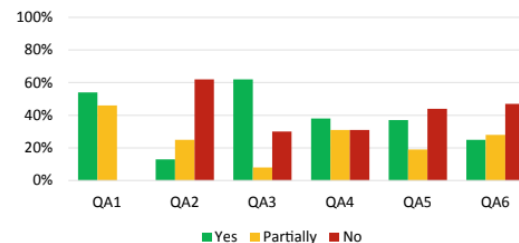
This is time-consuming, but it is an investment.

You will get a clear picture of your research and save time writing the SOTA!

Quality assessment questionnaire (optional)?

After reading each article, answer the following quality assessment questions (YES, PARTIALLY, NO) concerning:

- Purpose
- Evolve the state of the art
- Details on methodology
- Use of proven tools
- Application
- Limitations and future work discussion.



Example from [1]

QA1 Did the study consider the purpose and application domain of the considered metamodeling approach?

- Yes: it explains the purpose of metamodeling approach and its application domain
- Partially: it explains the purpose of metamodeling approach, but not the application domain
- No: it does not mention the purpose and application domain

QA2 Did the study extend the state-of-the-art in metamodeling?

- Yes: it extends the state-of-the-art in metamodeling
- Partially: it shows some new results in the metamodeling domain
- No: it applies an existing metamodeling approach to some domain

QA3 Did the study consider and give details on the methodology used?

- Yes: it clearly explains the methodology used
- Partially: it lists a few methods but not in a structural way
- No: it did not discuss the methodology used

QA4 Did the study use a proven meta-metamodel?

- Yes: it names the meta-metamodel and how it is used in the research
- Partially: it names the meta-metamodel but did not describe how the meta-metamodel is used
- No: it mentions the meta-metamodel name without explanation

QA5 Did the study consider an industrial use of the metamodeling approach?

- Yes: it discusses the use of metamodeling approach in the industry
- Partially: it mentions the possible use of metamodeling in the industry
- No: it does not mention the industrial use of metamodeling

QA6 Did the study consider the future work and discuss limitation?

- Yes: it explains the possible future elaboration of metamodeling approach and its limitations
- Partially: it mentions either future works or limitations
- No: it did not mention the limitations nor the possibility of future work

SLR Quality assessment questionnaire

The SLR QA allows the creation of a table and a corresponding discussion per topic (on the right, example table for limitations)

Table 14 The limitations of reviewed studies

Limitations description	References
The proposed metamodel did not support spatial information	[19]
The application of metamodeling is effective, but manual coding is still needed	[15]
The implementation of automation components depends on the metamodel process used to define the software process line	[62]
In-depth knowledge of UML is needed to apply metamodeling	[78]
The proposed metamodel can only support a limited system analysis	[52]
The metamodel is limited to the predefined requirement functionality	[53]
Limited possibility to support the changing learning algorithm	[5]
Limited usability of the proposed model editor	[76]
Cannot support large domain-specific systems	[26]
Open platform communications unified architecture is not aligned with UML	[20]
The proposed metamodeling strategy cannot be validated for large data	[43]

Example from [1]

Final Considerations

While you are reading the full papers, collect information!

Summarise main goals, contributions and limitations, etc., as well as annotate any other relevant information.

Be methodical! Be comprehensive!

Or do you want to repeat the process over and over because you missed a detail?

Final Considerations - citation management tools

You will want these features:

Collect information in an efficient way!

Mendeley, Zotero, EndNote?

It may require manual curation of references....

Protocol development
Database integration
Ease of import & export
Duplicate removal
Article screening
Critical appraisal
Assist with reporting
Meta-analysis

Final Considerations - manage literature reviews and evidence synthesis

- Elicit, Covidence, Rayyan, EPPI-Reviewer, DistillerSR...
- ConnectedPapers, ResearchRabbit...

Use of AI?

your responsibility

be transparent!

You will want these features:

Protocol development
Database integration
Ease of import & export
Duplicate removal
Article screening
Critical appraisal
Assist with reporting
Meta-analysis

Final Considerations

Collected data is the working ground for writing:

- Critical analysis: the end sections in the Literature Review chapter
- The State of the Art (another chapter)

Revisit (and improve) the context, problem and research question formulation.

References

[1] M. Fatehah, V. Mezhuyev, and M. Al-Emran, “*A Systematic Review of Metamodelling in Software Engineering*”, *Recent Advances in Intelligent Systems and Smart Applications*, vol. 295, pp. 3-27, Jun. 2020, doi: 10.1007/978-3-030-47411-9_1.

Example from:

[2] Oliveira, Patrícia, “*Graph-Based Visualisations for Time-Evolving Data: visualisation of the temporal evolution of players in national teams and clubs*”, Msc thesis, M.EIC, FEUP, 2024

Dissertation Preparation

• Questions?